



Department of Electrical and Electronic Engineering Hardware / Software(✓) Report-2 Submission

Semester: Spring 26

Course Code: EEE283L

Course Title: Digital Logic Design

Section:03

Experiment No:03

Experiment Name: Study and Application of Combinational Circuit Design
Blocks: Multiplexer and Demultiplexer.

Group No: 02

Group members:

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Prepared by:

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Name of the Experiment: Study and Application of Combinational Circuit Design
Blocks: Multiplexer and Demultiplexer.

Introduction:

Multiplexer (MUX) and Demultiplexer (DEMUX) are important combinational logic blocks used in digital systems. A multiplexer selects one input from many inputs and sends it to a single output line while a demultiplexer takes one input and sends it to one of many output lines. In this experiment, MUX and DEMUX are designed using fundamental logic gates and also verified using IC and Proteus simulation to understand their working and applications.

Objective:

The main objective of this experiment is to study and implement Multiplexer and Demultiplexer circuits using fundamental logic gates. It also aims to verify the operation of MUX and DEMUX ICs and implement a given logic function using a 8-to-1 multiplexer.

Software:

- Proteus & Professional
- Digital logic library components inside Proteus

Components:

- Subcircuit component
- Logic Gates (AND, OR, NOT)
- Logic toggle
- Logic probe
- LED indicators
- 8 to 1 MUX IC (74151)
- Connecting wires

Simulation Diagram:

2 To 1 MUX

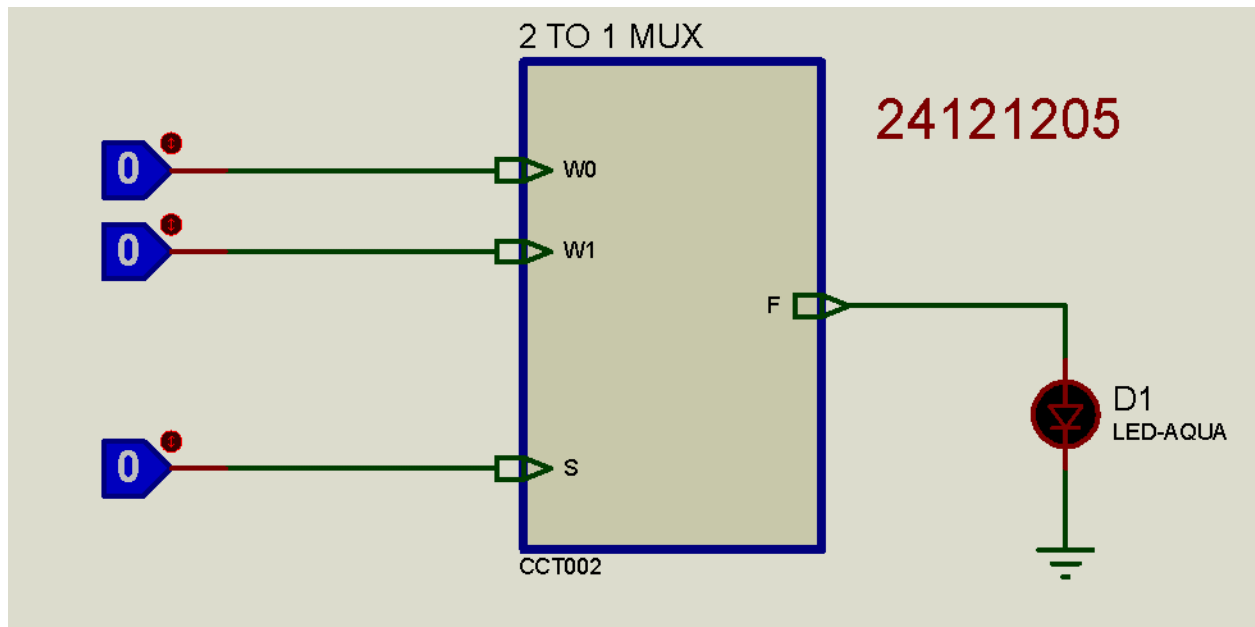


Figure:- Diagram of a 2-to-1 MUX subcircuit block

Child Sheet

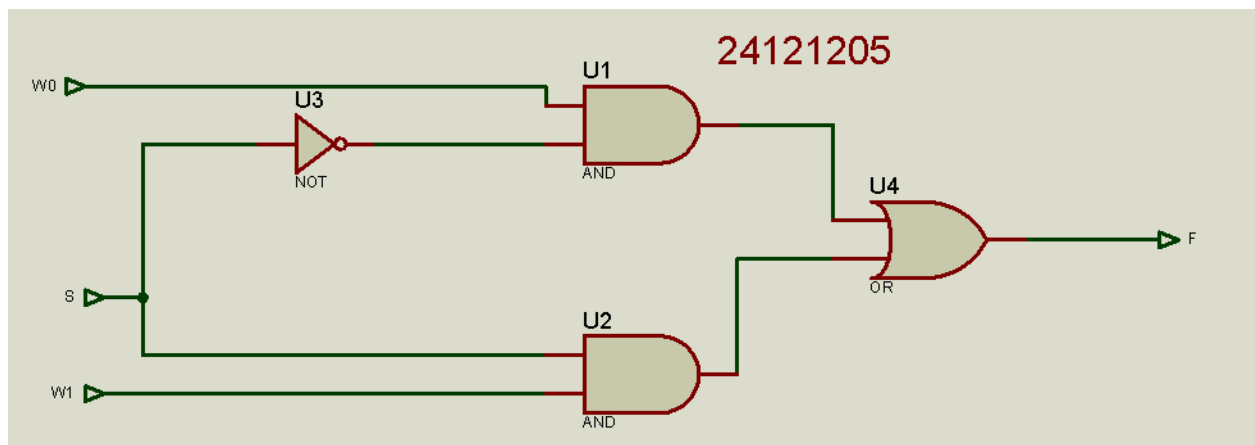


Figure:- Internal logic gate implementation (Child Sheet) of the 2-to-1 MUX

Test Result:-

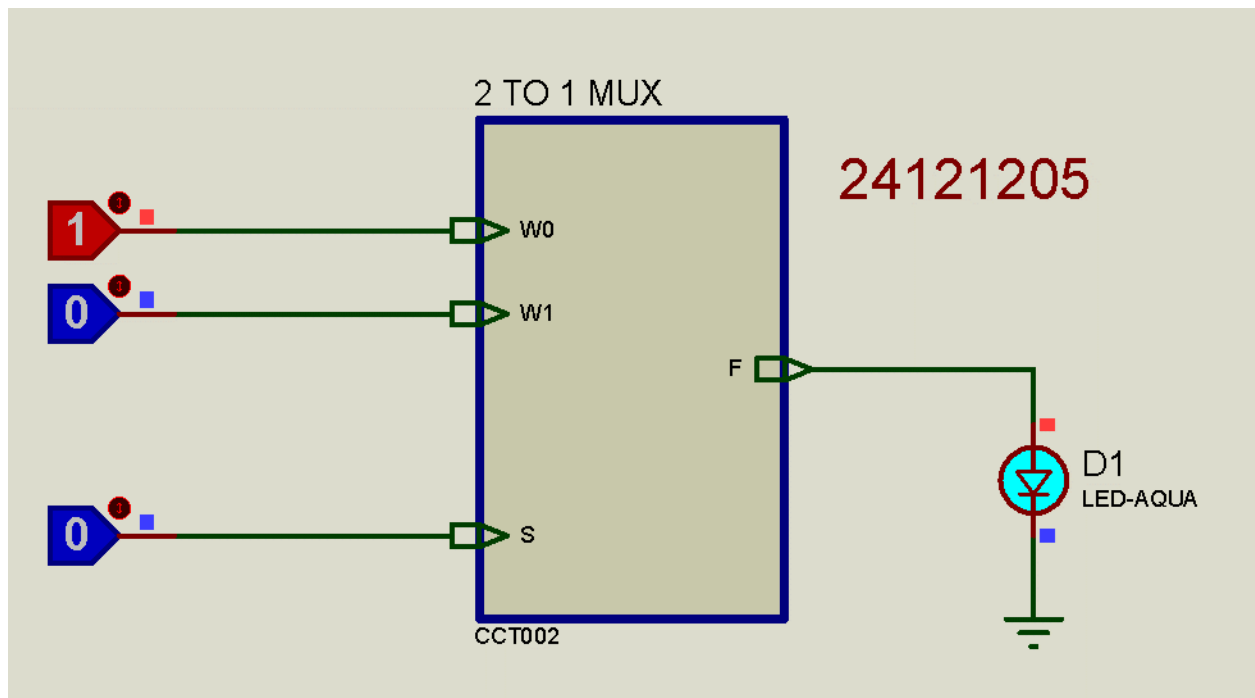


Figure:- Functional verification of 2-to-1 MUX for S=0, Y=W0

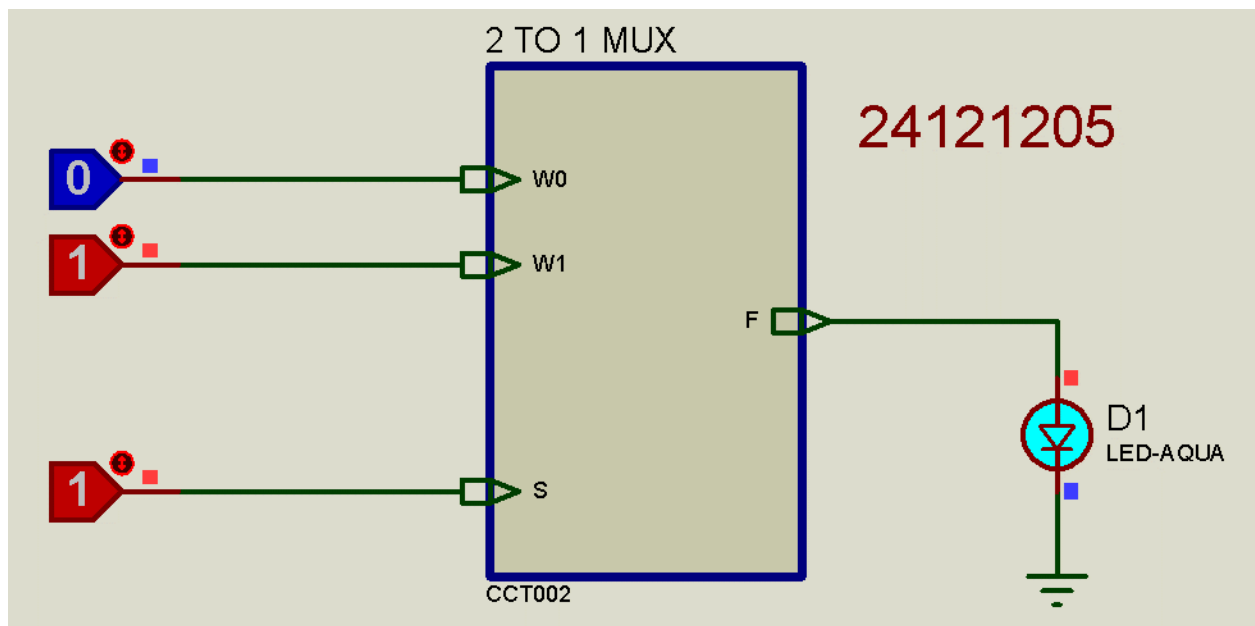


Figure:- Functional verification of 2-to-1 MUX for S=1, Y=W1

Result (Truth Table):-

Equation: ($\bar{S}W0 + SW1$)

S	F
0	W0
1	W1

Simulation Diagram:

1 To 2 DEMUX

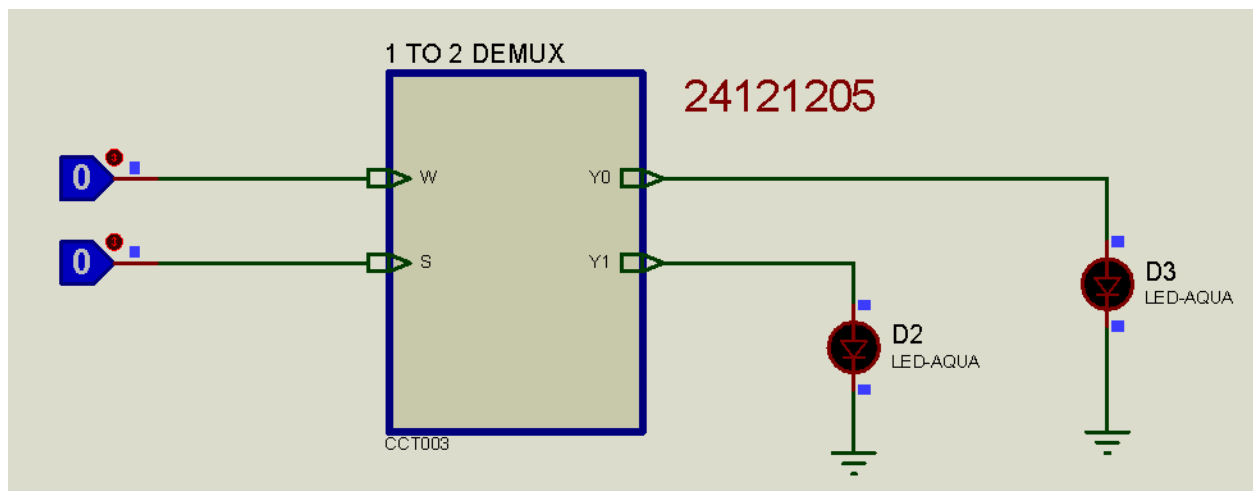


Figure:- Diagram of a 1-to-2 DEMUX subcircuit block

Child Sheet

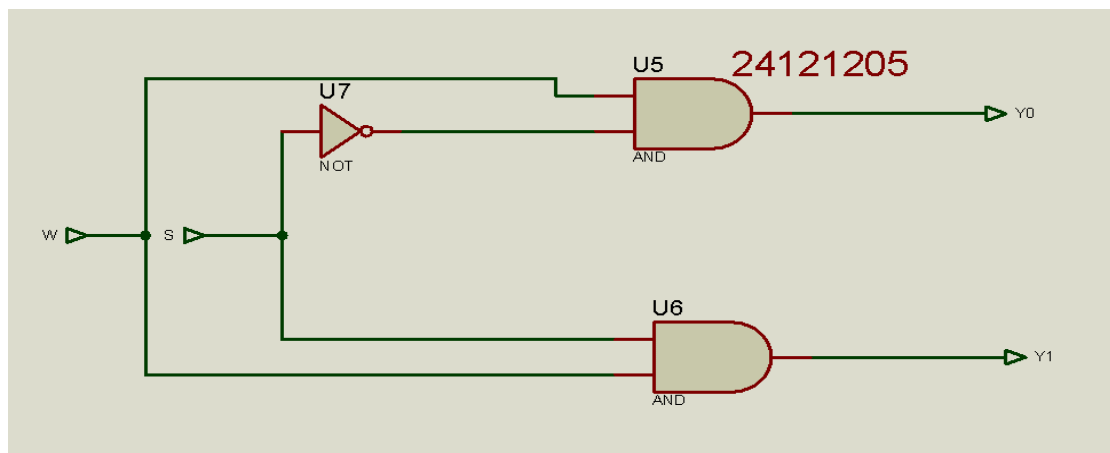


Figure:- Internal logic gate implementation (Child Sheet) of the 1-to-2 DEMUX

Test Result:-

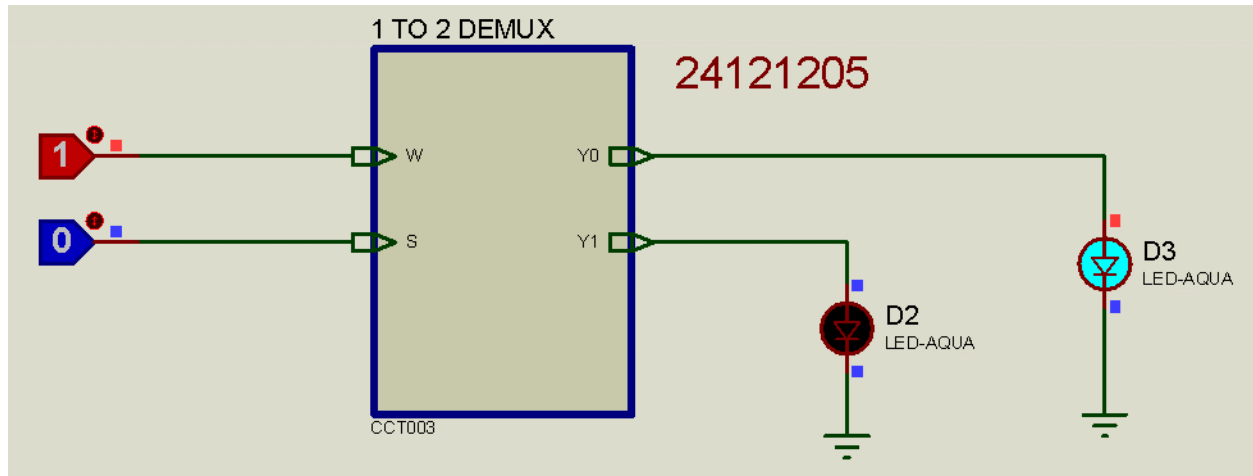


Figure:- Logic state showing Output Y0 active (S = 0, W = 1).

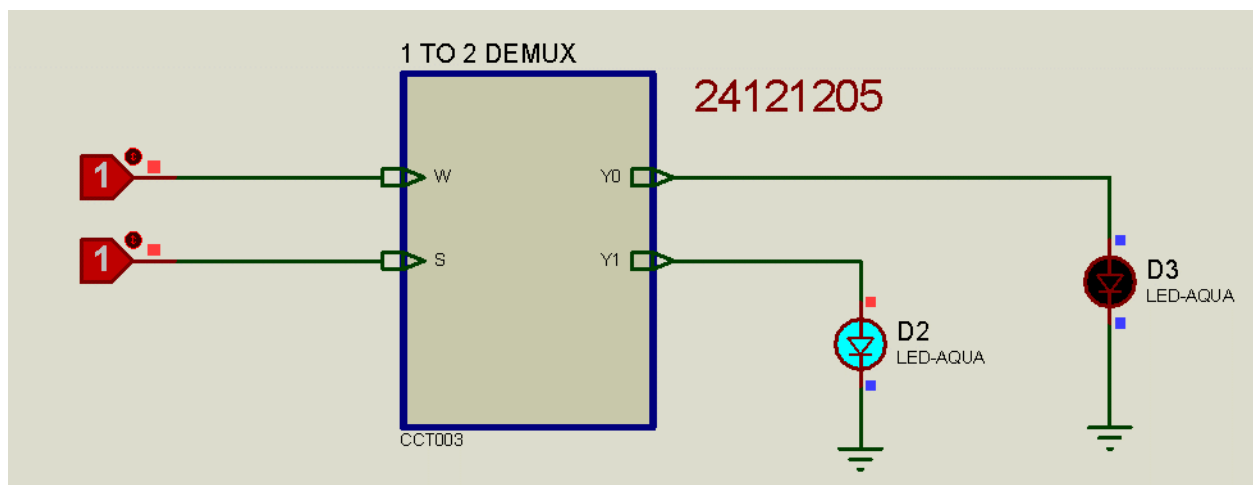


Figure:- Logic state showing Output Y1 active (S = 1, W = 1).

Result (Truth Table):-

Equation: $(Y_0 = \bar{S} \cdot w), (Y_1 = S \cdot W)$

S	W	Y_0	Y_1
0	0	0	0
0	1	1	0
1	0	0	0
1	1	0	1

Simulation Diagram:

4 To 1 MUX

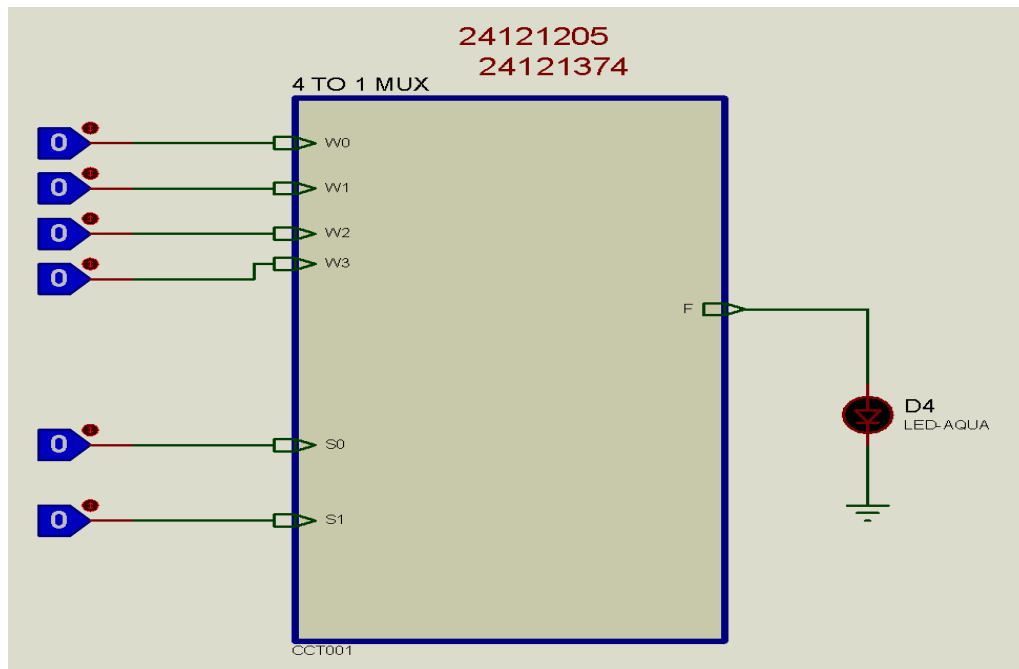


Figure:- Diagram of a 4-to-1 MUX subcircuit block

Child Sheet

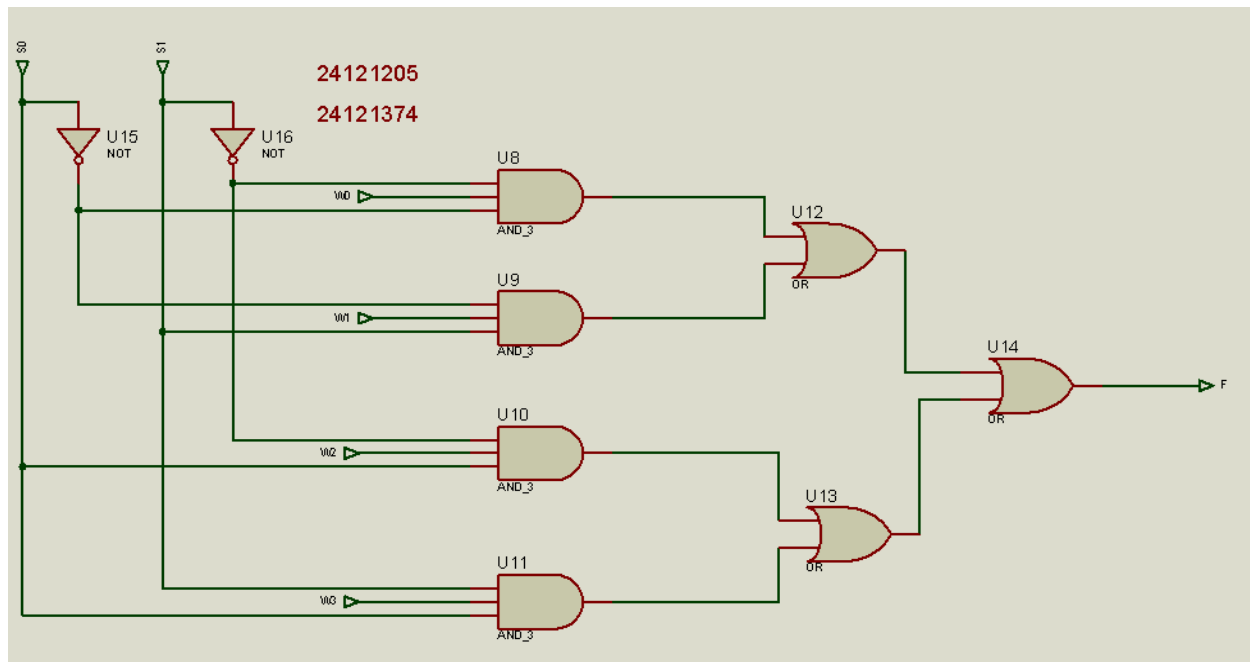


Figure:-Internal logic gate implementation (Child Sheet) of the 4-to-1 MUX

Test Result:-

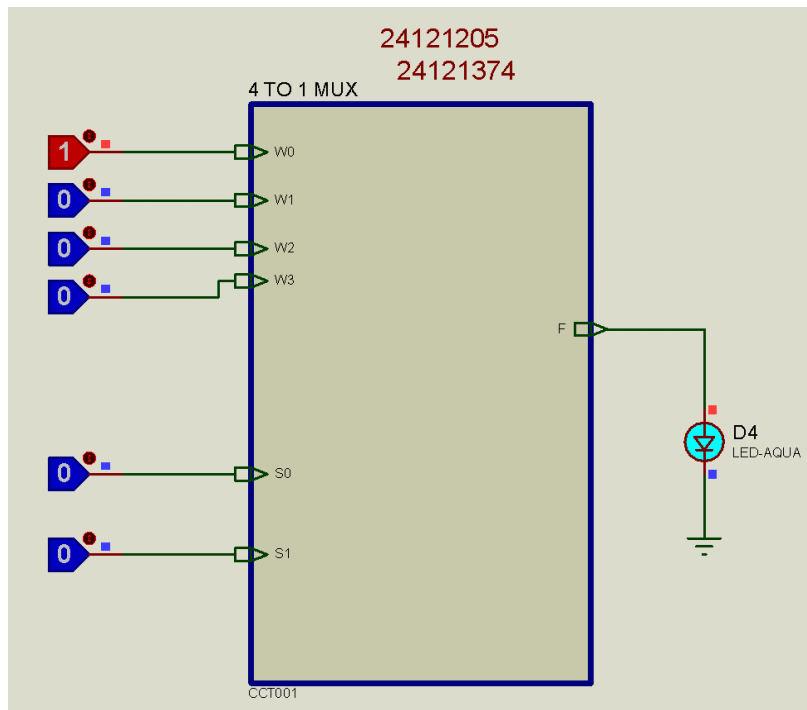


Figure:- MUX selecting Input **W0** (S1=0, S0=0).

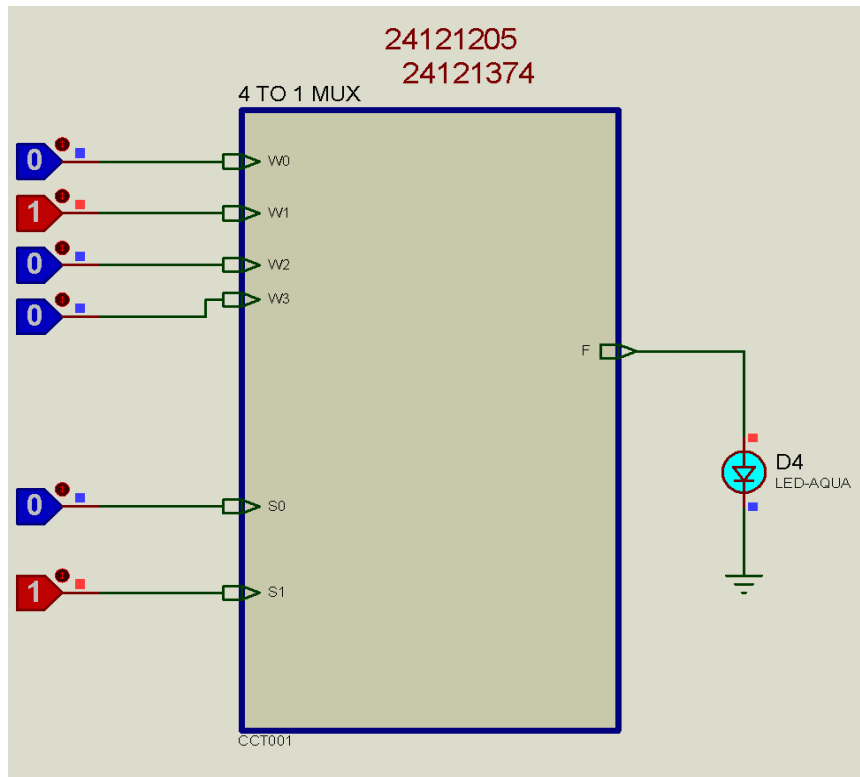


Figure:- MUX selecting Input **W1** (S1=0, S0=1).

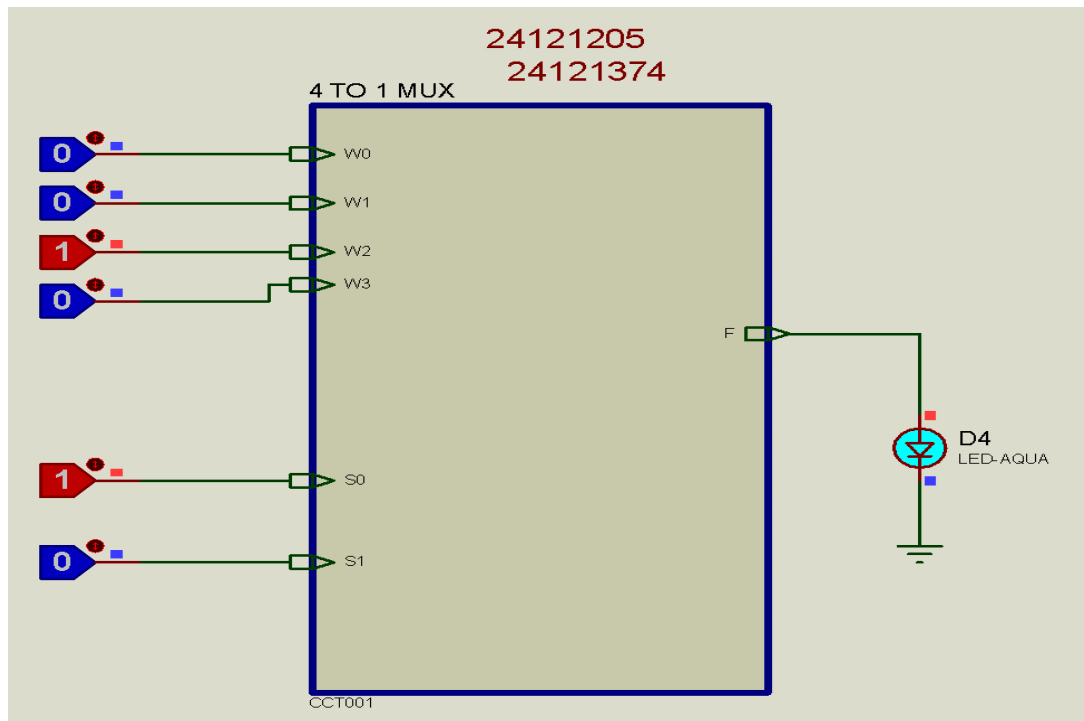


Figure:- MUX selecting Input **W2** ($S1=1$, $S0=0$).

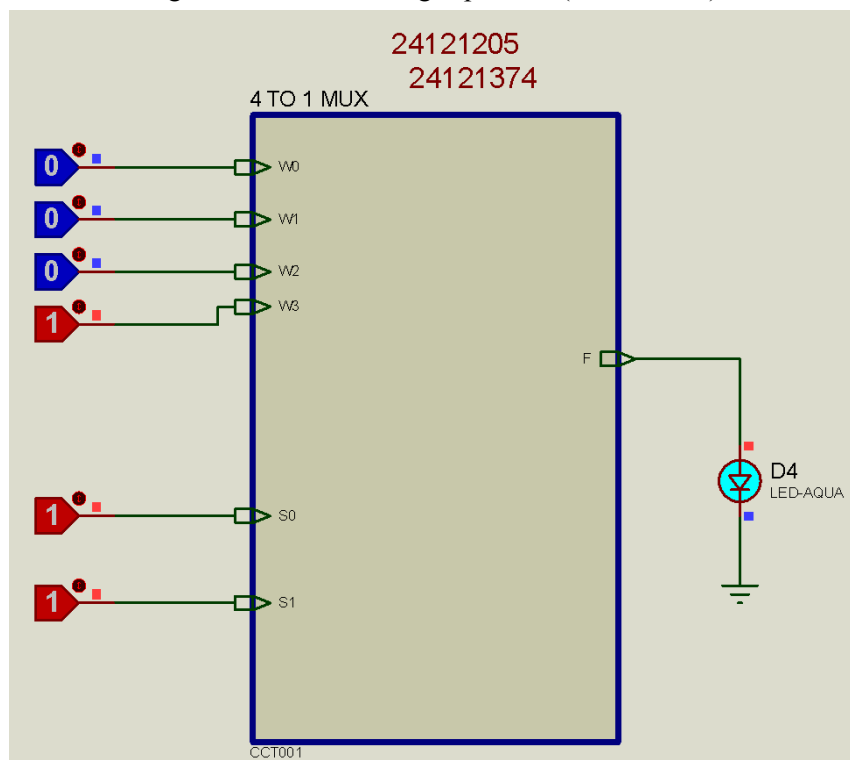


Figure:- MUX selecting Input **W3** ($S1=1$, $S0=1$)

Result (Truth Table):-

$$\text{Equation: } F = (\overline{S1} \cdot \overline{S0} \cdot W0) + (S1 \cdot \overline{S0} \cdot W1) + (\overline{S1} \cdot S0 \cdot W2) + (S1 \cdot S0 \cdot W3)$$

S1	S0	F
0	0	W0
1	0	W1
0	1	W2
1	1	W3

Simulation Diagram:

1 To 4 DEMUX

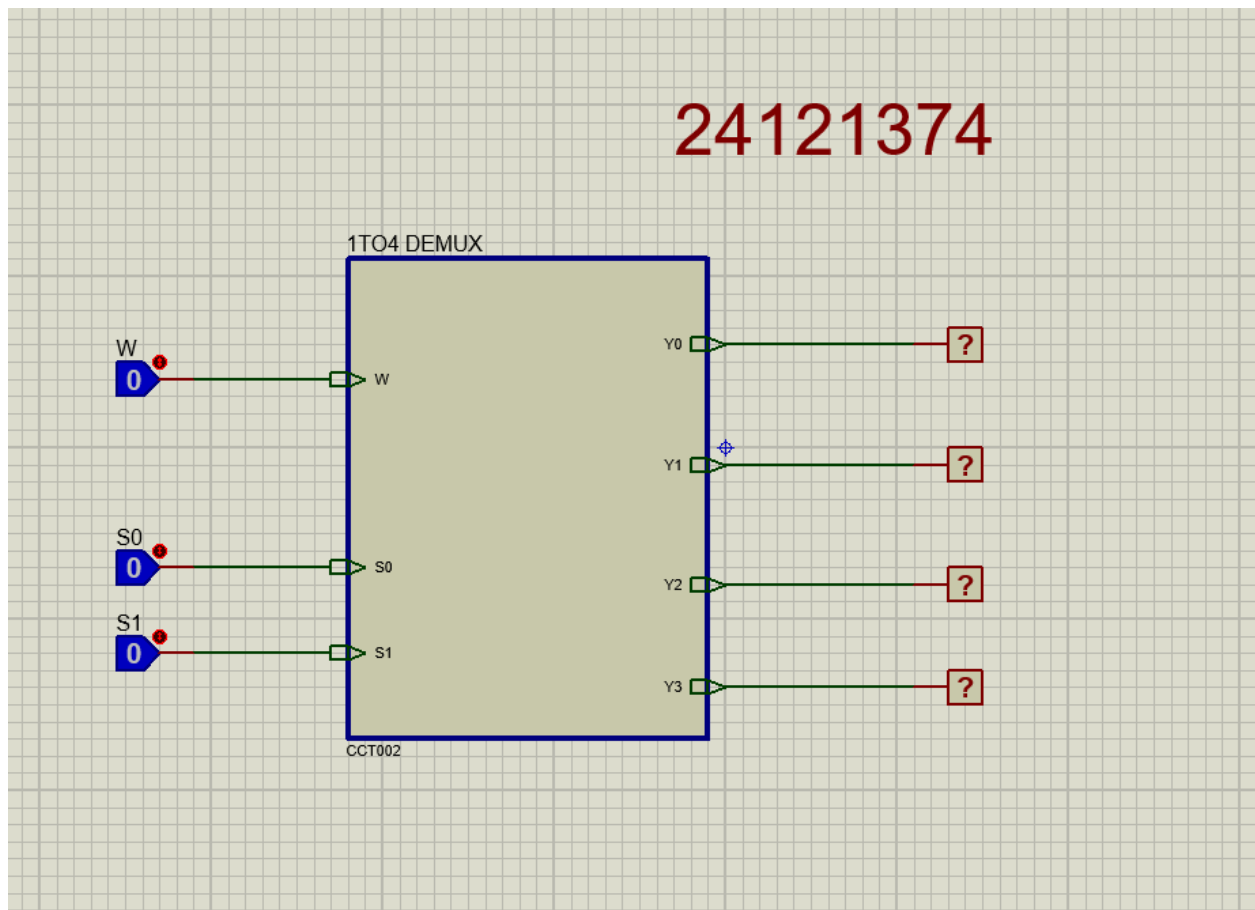


Figure: Diagram of a 1-to-4 DEMUX subcircuit block

Child Sheet:

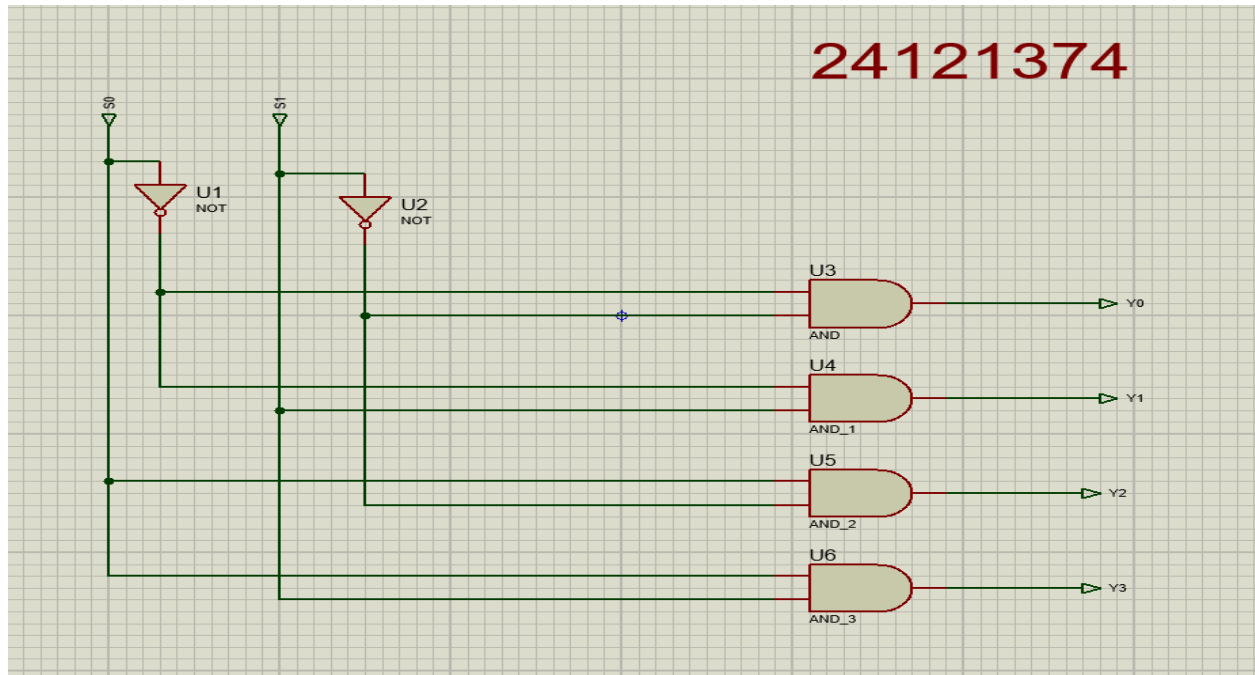


Figure: Internal logic gate implementation (Child Sheet) of the 1-to-4 DEMUX

Test Result:

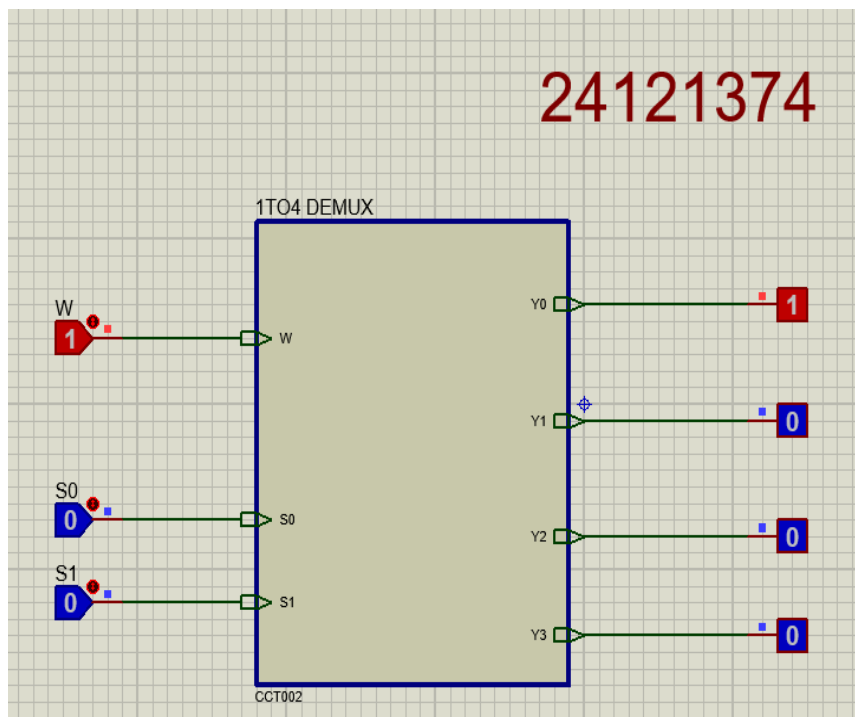


Figure: Logic state showing output Y0 active (S0=0, S1=0)

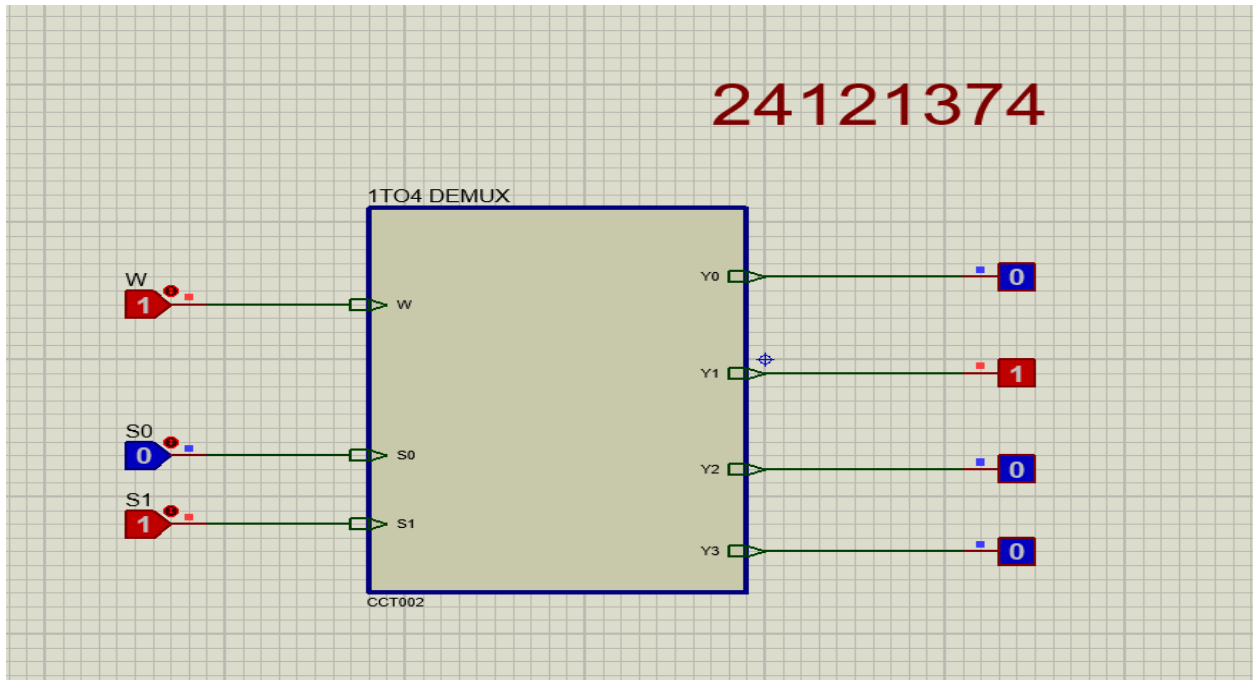


Figure: Logic state showing output Y1 active (S0=0, S1=1)

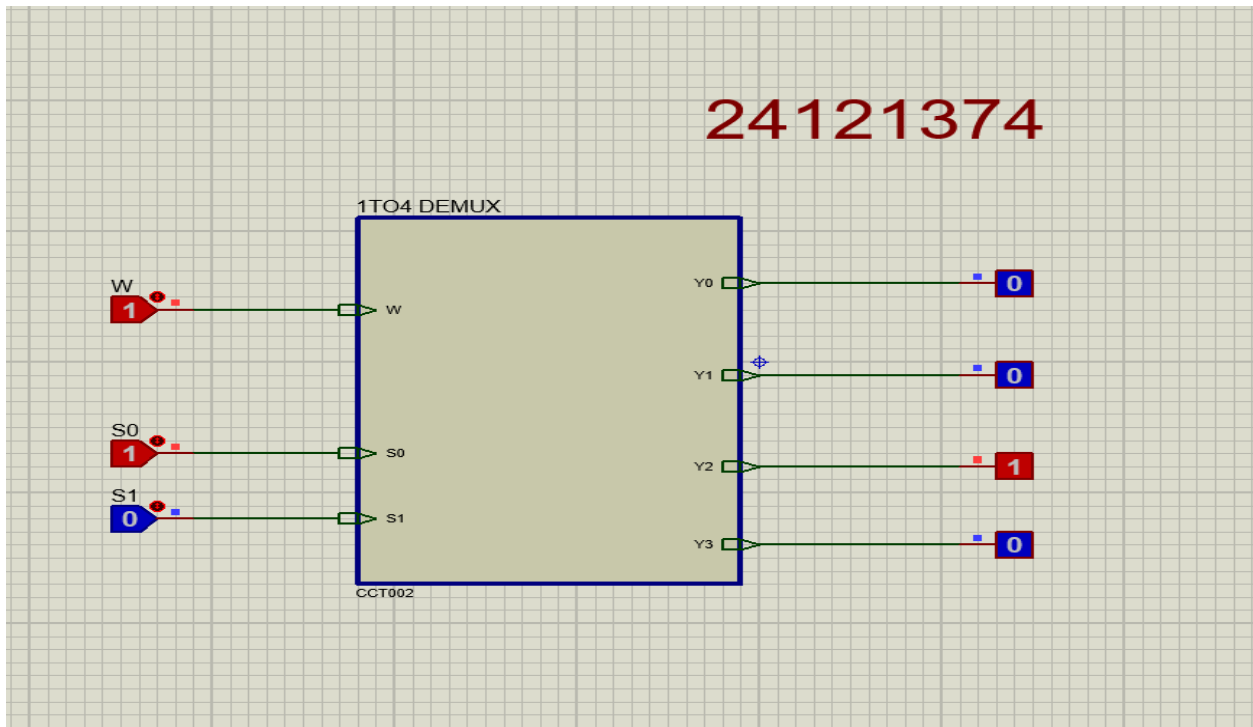


Figure: Logic state showing output Y2 active (S0=1, S1=0)

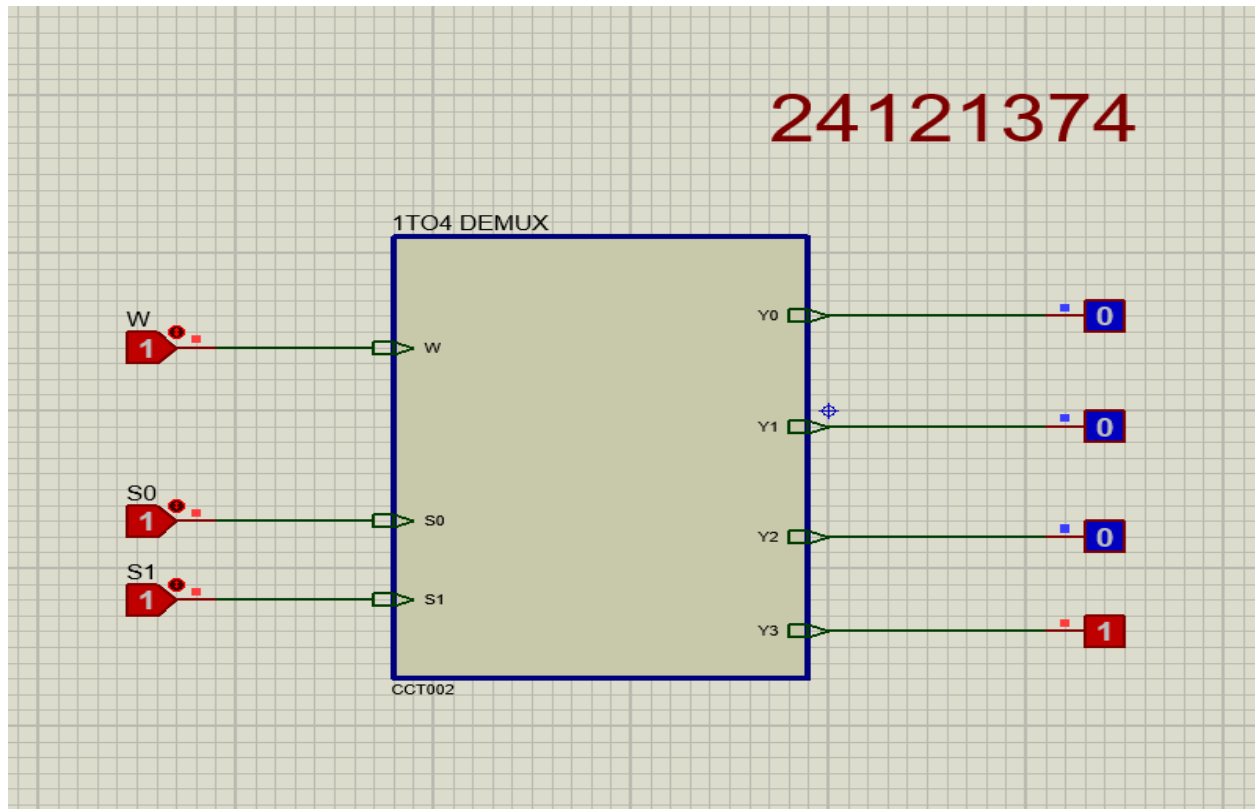


Figure: logic State showing output Y3 active (S0=1, S1=1)

Result (Truth Table):

Equation: $(Y0 = \overline{S0} \cdot \overline{S1})$, $(Y1 = \overline{S0} \cdot S1)$, $(Y2 = S0 \cdot \overline{S1})$, $(Y3 = S0 \cdot S1)$

Input	Selection Line		Output			
W	S0	S1	Y0	Y1	Y2	Y3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	0
0	X	X	0	0	0	0

Simulation Diagram:

Using 8 to 1 MUX IC (74151); $F(A,B,C)=\sum m(1,3,6,7)$

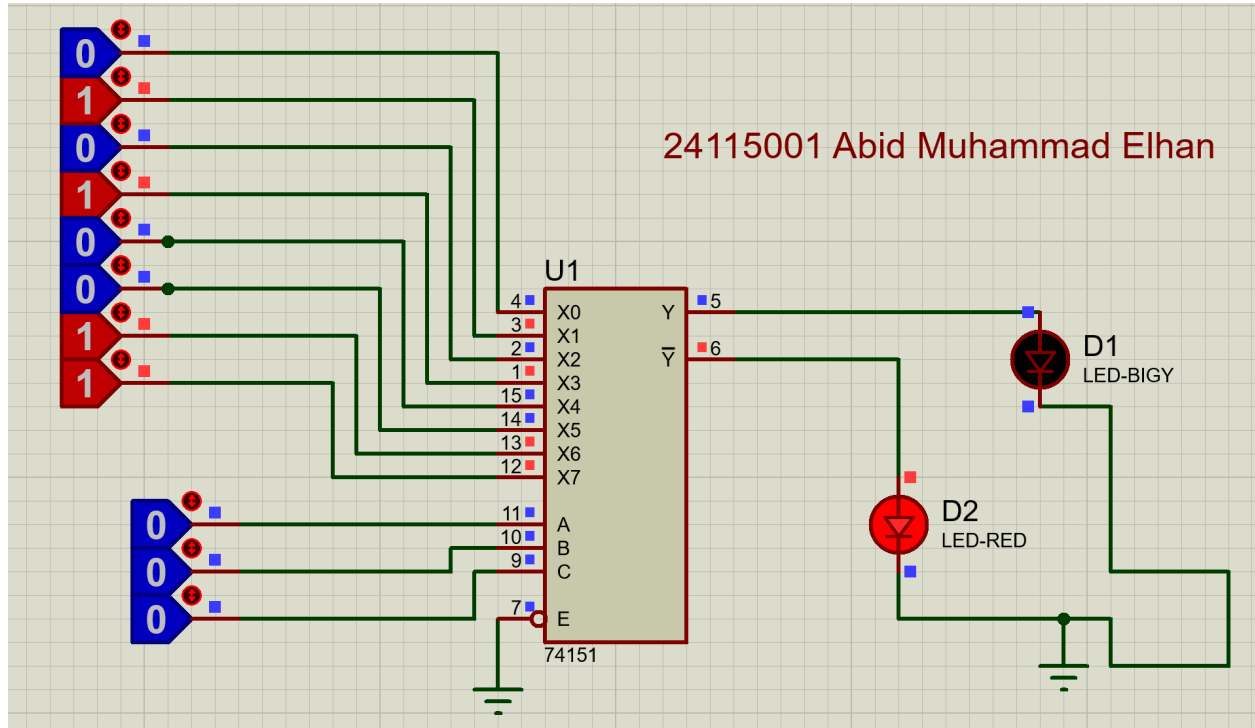


Figure:- Diagram of a 8-to-1 MUX (74151)

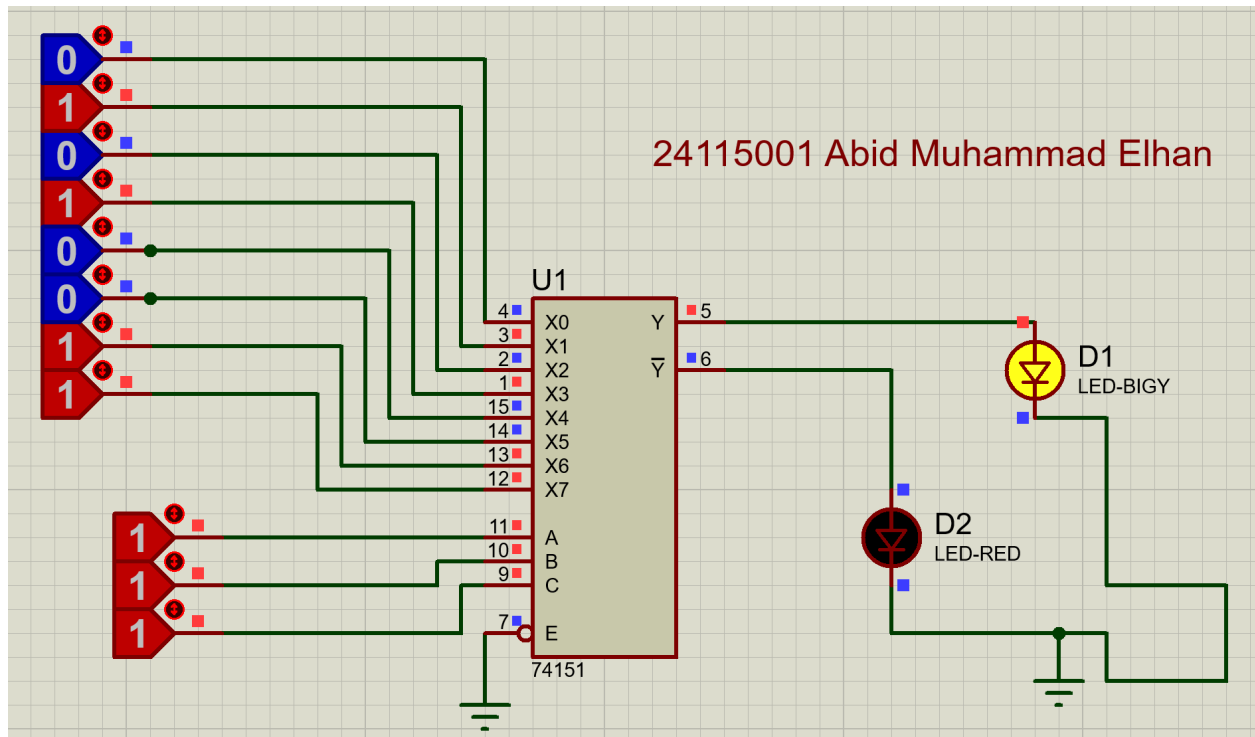


Figure 2:- Diagram of a 8-to-1 MUX (74151)

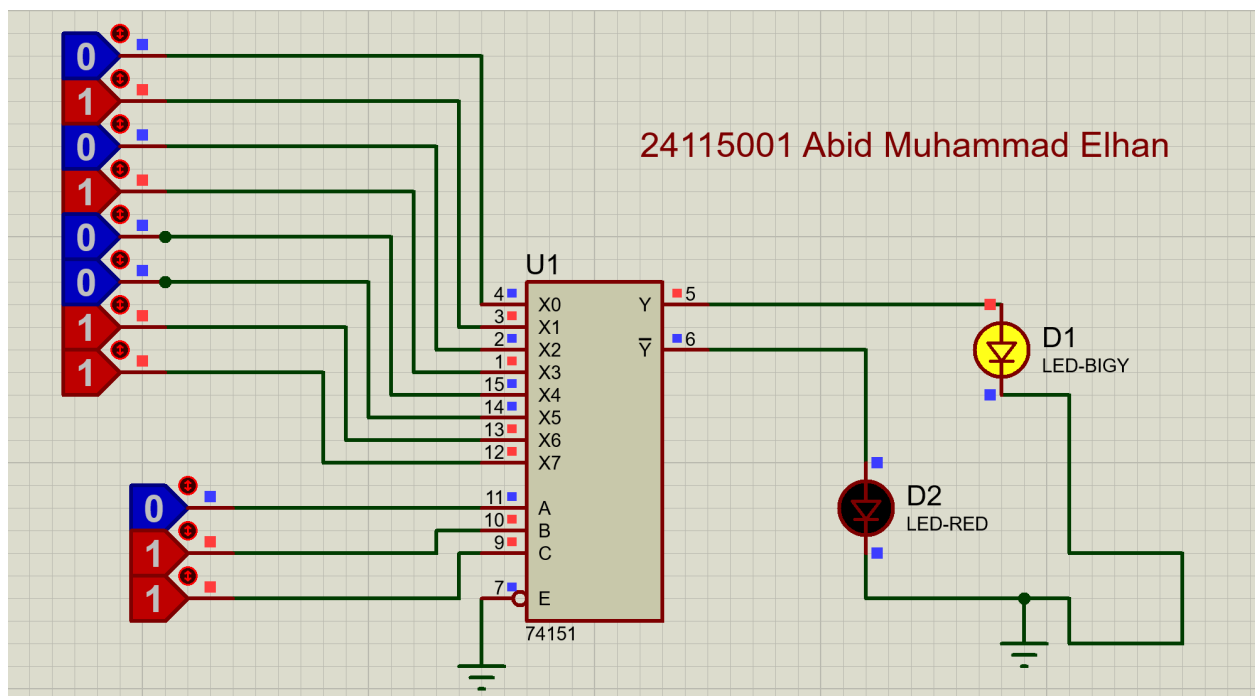


Figure 3:- Diagram of a 8-to-1 MUX (74151)

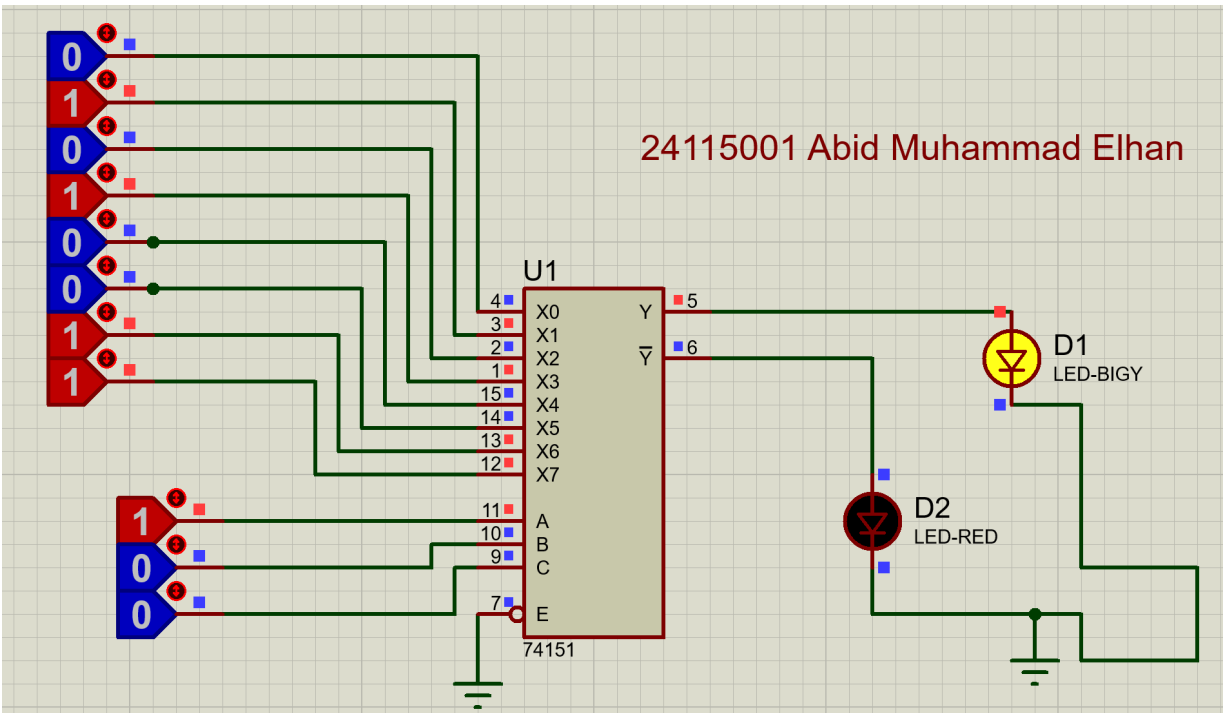


Figure 4:- Diagram of a 8-to-1 MUX (74151)

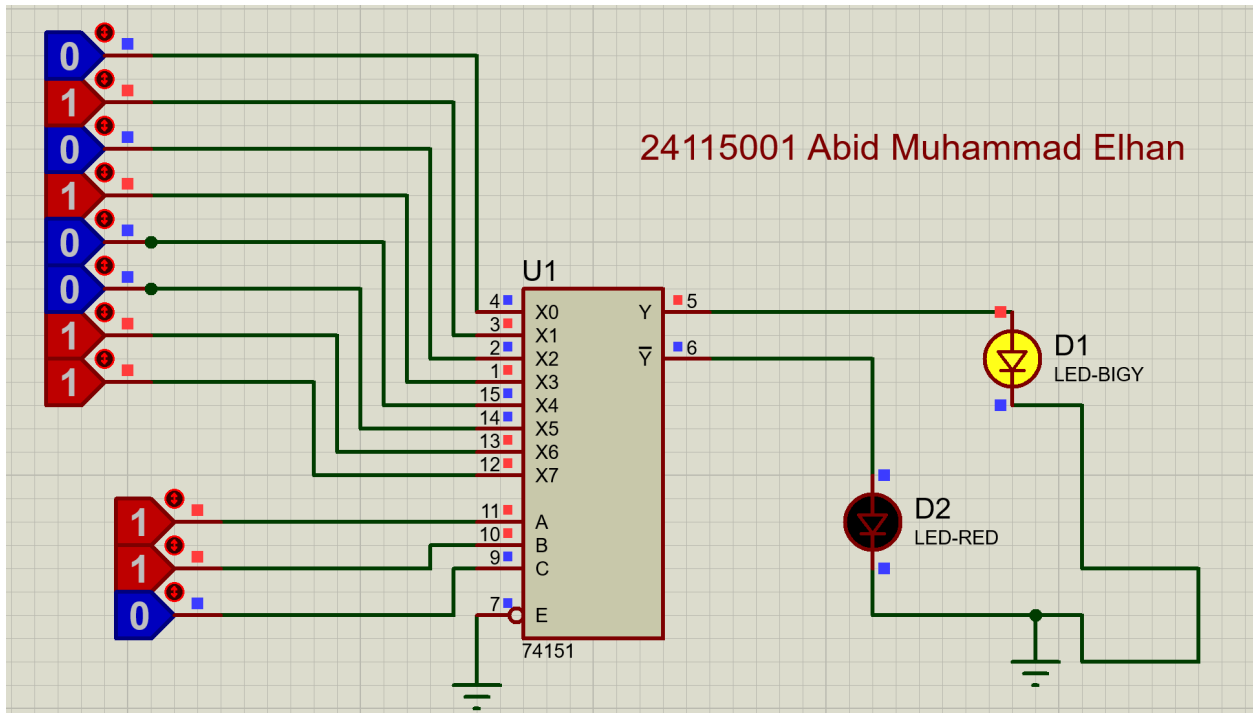


Figure 5:- Diagram of a 8-to-1 MUX (74151)

Results:

Row	A	B	C	F	I
1	0	0	0	0	I0
2	1	0	0	1	I1
3	0	1	0	0	I2
4	1	1	0	1	I3
5	0	0	1	0	I4
6	1	0	1	0	I5
7	0	1	1	1	I6
8	1	1	1	1	I7

Discussion:-

In this experiment, we designed different combinational circuits like 2-to-1, 4-to-1 Multiplexer and 1-to-2, 1-to-4 Demultiplexer using logic gates in Proteus software. We also verified the working of the 8-to-1 MUX IC (74151). From the simulation results and truth tables, we clearly observed that the multiplexer selects one input based on the selection line and sends it to the output. Similarly, the demultiplexer sends one input signal to the selected output line. All the outputs matched with the theoretical equations and truth tables.

Conclusion:-

This experiment improved our understanding of combinational circuit design. We successfully verified the operation of MUX and DEMUX circuits both theoretically and practically.

Proteus File Google Drive link:

<https://drive.google.com/drive/u/5/folders/19g-u2tSShNyCv08okYAd-Ha-BjZ0PwBf>